

MSDS 7333: Case Study 3
Spam and Ham Email Classification Analysis

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Chapter 1

Introduction

Email spam, also known as junk email, refers to unsolicited messages sent in bulk via electronic messaging systems. Because of these unwanted emails, there is a need to segregate legitimate correspondence (ham) from unwanted solicitations (spam). This discrimination task represents a canonical binary classification problem in text analysis and information security domains.

This study addresses the automated classification of electronic mail communications into spam and ham categories based on textual content analysis. The classification system must exhibit high accuracy while minimizing both false positive identifications (legitimate emails incorrectly flagged as spam) and false negative identifications (spam incorrectly classified as legitimate).

The input data utilized in this case study comprises a collection of email communications extracted from various sources, each pre-labeled as prefix of either spam or ham. These communications contain diverse textual features including: email headers containing metadata (sender, recipient, date, subject), message body content in various formats (plain text, HTML, multipart). These messages may contain potentially embedded URLs, imagery references, and attachments and has varying degrees of message complexity and length

The primary objectives of this investigation include:

1. Construction of a solid methodology for email content extraction and classification to corresponding target class, output is saved to separated csv file.
2. Identification of discriminative language features between spam and ham communications
3. Develop a Machine Learning models for spam email classification using Naive Bayes.

4. Quantitative assessment of classification performance metrics
5. Visualization of the results

The subsequent chapters detail the methodological approach, exploratory analysis, model development, and conclusions derived from this study.

Chapter 2

Data

The original data is a collection of email files in the `data/` directory. The files are organized into five directories: `easy_ham`, `easy_ham_2`, `hard_ham`, `spam`, `spam_2`. There are two main types of email messages: multipart and singlepart. While singlepart messages contain only one part and can be text/plain or text/html, multipart messages as their names suggested, have multiple parts, each part can be a different type of content (text, html, etc.). In order to read multipart messages, we need to loop through all the parts of the message and check the content type of each part before assigning the correct way to read it.

This chapter delineates the methodological approach employed to transform raw email files into a structured dataset and its corresponding label, which is amenable to statistical analysis and machine learning applications. The initial phase involved systematic traversal of the directory structure to identify and categorize email files. We construct a code to systematically crawl the `data/` directory and its subdirectories, then identifies files as either spam or ham based on path context (email in folder containing `spam` is spam, otherwise ham) After that, we assigns binary labels (0 for ham, 1 for spam) and records file paths for subsequent content extraction

The extraction of textual content from email files presented significant challenges due to the heterogeneous nature of email formats. The preliminary analysis identified different content types required specialized processing methods:

```
Counter({'text/plain': 7413,  
        'multipart/alternative': 326,  
        'text/html': 1193,  
        'multipart/mixed': 179,  
        'multipart/related': 56,
```

```

'multipart/signed': 180,
'text/plain charset=us-ascii': 1,
'multipart/report': 5})

```

in which: `text/plain`, `'text/plain charset=us-ascii'` are single part text files and could be extracted by `get_payload` command, while `text/html` is another singlepart HTML format and can be extracted using `BeautifulSoup` library from `get_payload` output. For multipart related messages: we build a function called `get_multipart` approach to extract the content in the nested structure. The following snippet shows the code for these functions:

```

def get_txt_from_type(type_, msg):
    if type_ == "text/plain":
        txt = msg.get_payload()
    elif type_ == "text/plain charset=us-ascii":
        txt = msg.get_payload()
    elif type_ == "text/html":
        soup = BeautifulSoup(msg.get_payload(decode=True), 'html.parser')
        txt = soup.get_text(separator="\n", strip=True)
    else:
        txt = get_multipart(msg)
    return txt

def get_multipart(msg):
    payloads = msg.get_payload()
    content_out = ""
    if type(payloads) == list:
        for part in payloads:
            content_type = part.get_content_type()
            content_disposition = part.get_content_disposition()
            charset = part.get_content_charset() or 'utf-8'

            if content_type == 'text/plain' and content_disposition is None:
                content1 = part.get_payload(decode=True).decode(charset)
                content_out=content_out+content1
            elif content_type == 'text/html' and content_disposition is None:
                soup = BeautifulSoup(part.get_payload(decode=True), 'html.parser')
                content2 = soup.get_text(separator="\n", strip=True)
                content_out=content_out+content2
    else:
        soup = BeautifulSoup(msg.get_payload(decode=True), 'html.parser')

```

```

        content_out = soup.get_text(separator="\n", strip=True)

    return content_out

```

This approach enabled the direct extraction of plain text content; ensure HTML parsing with BeautifulSoup to extract readable text and recursive extraction from multipart messages. The extraction methodology was applied to the entire corpus using a systematic batch approach using following code:

```

ftext = []
for item in contents:
    with open(item, "r", encoding="latin-1") as f:
        msg = email.message_from_file(f)
        type_ = msg.get_content_type()
        txt = get_txt_from_type(type_, msg)
        ftext.append(txt)

```

This implementation opens each file using Latin-1 encoding to accommodate various character sets, parses the email structure using Python's built-in email module, determines the content type for appropriate processing and extracts and stores the textual content.

The final phase involved the assembly of a structured dataset by converting to pandas dataframe and saving to csv file. The created a tabular dataset has filename identifiers for traceability, extracted textual content for analysis and binary classification labels (0=ham, 1=spam).

The methodology detailed in this chapter established a robust pipeline for transforming heterogeneous email files into a structured dataset, addressing the complexities inherent in email communications. The resulting **emails.csv** file provides a foundation for subsequent exploratory data analysis and classification model development.

Chapter 3

Exploratory Data Analysis

This chapter presents a comprehensive exploratory analysis of the email dataset generated through the methodological approach described in Chapter 2. The analysis examines multiple dimensions of the corpus to elucidate distinguishing characteristics between spam and legitimate communications.

The initial examination focused on fundamental dataset properties such as dimension and target labels: There are 9353 emails in the dataset with the target labels are 0 for ham and 1 for spam. The class distribution is 75% ham and 25% spam as shown in Figure 3.1.

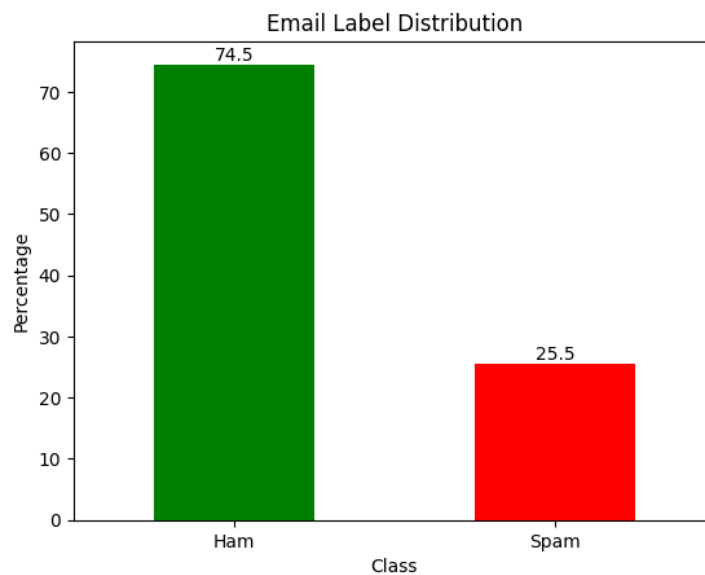


Figure 3.1: Distribution of ham and spam emails in the dataset

From our preliminary analysis, there are 146 emails out of 9353 which

didn't have any value (NaN), it might be due several reasons: blank emails, contain only images with links to some harmful sources, could not be converted from original email or they are not emails, but some other files. Since the objective of our Case Study is to do the modeling and the number of missing emails are small (1.5%), we remove them for now.

A fundamental distinguishing characteristic between spam and legitimate communications is message length. Based on the initial dataset, we create a new column called `word_count` which is the count of the total number of words in the corresponding email content. We split the collection of emails into two groups: ham and spam. The analysis revealed distinctive patterns in message length distributions between classes (Figure 3.2): spam messages exhibited higher average length (mean: 352 words); while ham messages showed narrower length variance of 274 words. Both distributions displayed right skew, with many short messages and fewer very long ones. These length distribution characteristics suggest message length could serve as a useful feature for classification.

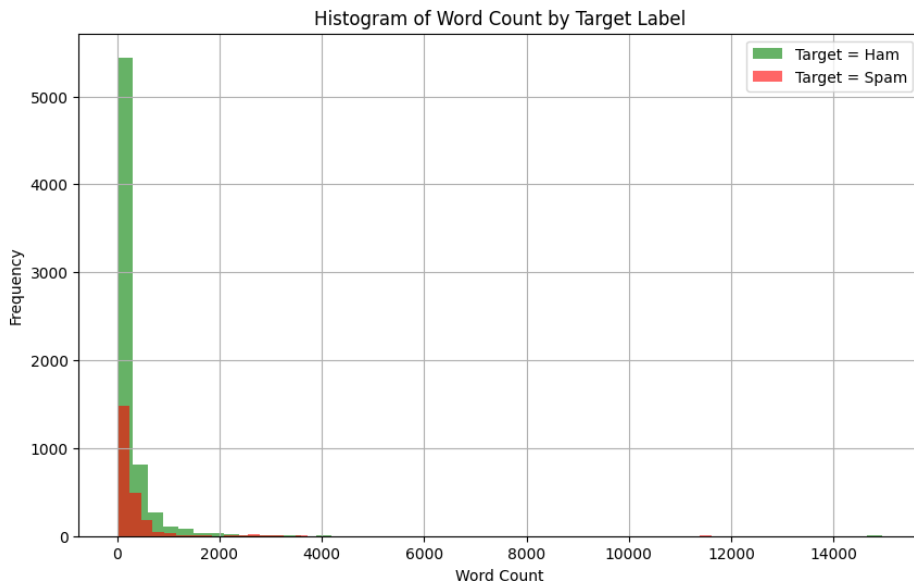
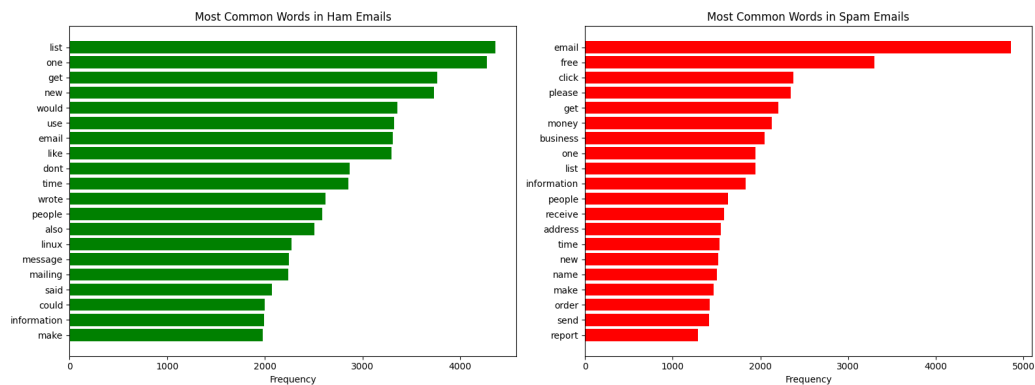


Figure 3.2: Distribution of email lengths by class

In addition to the number of words in the email content, we also count the frequency of each word in the email content. Finally, we visualize the most common words in each class (Figure 3.3).



The analysis revealed distinctive lexical characteristics such as ham communications featured more professional and technical vocabulary, spam messages exhibited high frequency of marketing and action-oriented terms and pronoun usage differed significantly between classes

Aggregate lexical patterns were visualized using word clouds to highlight dominant terminology (Figure 3.4):



The word cloud visualizations reinforced the lexical distinctions observed in frequency analysis while providing additional context regarding semantic domains present in each class.

Communication provenance can provide discriminative signals for classification. Domain analysis was conducted to identify the most frequent patterns for spam and ham emails (Figure 3.5):

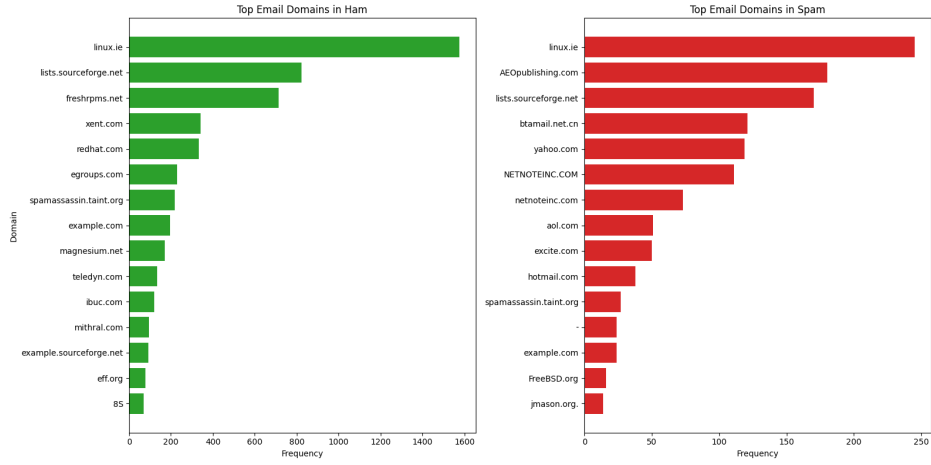


Figure 3.5: Most common email domains in the dataset

The domain analysis revealed that high prevalence of free email provider domains in spam communications, correlation between domain reputation and message classification and institutional domains strongly associated with legitimate communications

In conclusion, the exploratory analysis yielded several significant insights:

1. Message length characteristics differ substantially between classes
2. Lexical composition provides strong discriminative signals
3. Approximately 1.5% of records contain missing or invalid content
4. Domain patterns correlate with message classification

These findings inform feature engineering strategies for the subsequent modeling phase, suggesting emphasis on lexical features and appropriate handling of missing values.

Chapter 4

Modeling & Results

This chapter details the development and evaluation of statistical models for automated classification of email communications into spam and ham categories. The modeling approach employs Naive Bayes classification, a probabilistic technique particularly well-suited for text classification tasks due to its computational efficiency and effectiveness with high-dimensional sparse feature spaces. Prior to model development, several preprocessing steps were implemented to structure the data appropriately: To facilitate unbiased model evaluation, the dataset was partitioned into training (80%) and testing (20%) subsets. The stratified sampling approach ensures that the class distribution is maintained in both training and testing subsets, preventing sampling bias that could distort evaluation metrics. Next, we need to transform unstructured text into numerical features using the Term Frequency-Inverse Document Frequency (TF-IDF) approach. This TF-IDF method captures both unigram and bigram features to model contextual patterns and limits feature space to 5,000 most informative features to prevent overfitting.

The Multinomial Naive Bayes classifier was selected as the primary modeling approach which match the prior distribution of the input tokenized text with its corresponding target class for posterior probability. K-Fold cross-validation was implemented to assess model stability across different data partitions. Details of the cross-validation performance metrics are shown in Table 4.1.

Table 4.1: Cross-validation performance metrics

Metric	Value	Standard Deviation
Accuracy	0.982	0.0112

The final model was evaluated on the held-out test set to assess generalization performance and tabulated in Table 4.2.

Table 4.2: Test set performance metrics

Class	Precision	Recall	F1-Score	Support
Ham (0)	0.99	0.99	0.99	1358
Spam (1)	0.97	0.96	0.97	484
Accuracy		0.98		1842

The ROC curve is shown in Figure 4.1 with very high AUC score of 0.99.

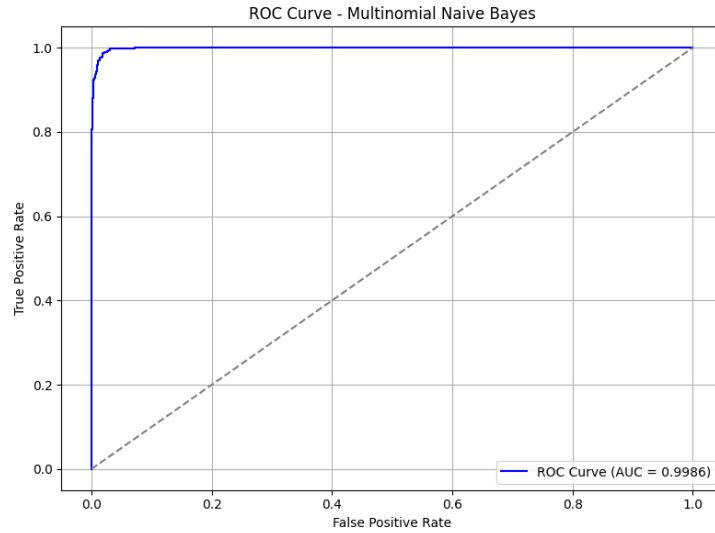


Figure 4.1: ROC curve for test set predictions

The Naive Bayes model facilitates interpretation of feature importance through examination of log-probability ratios. Top 20 most difference words between spam and ham domain are shown in Figure 4.2 and Figure 4.3.

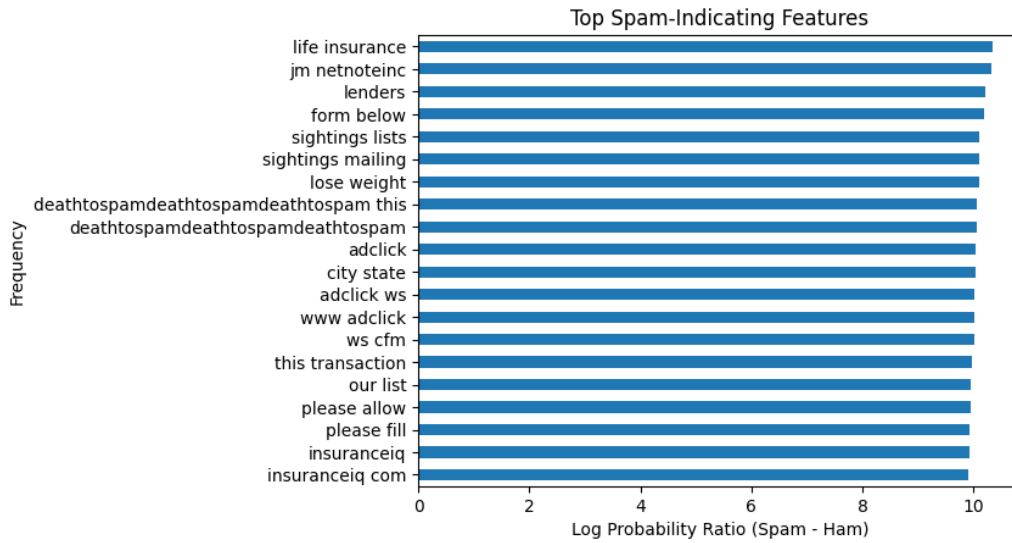


Figure 4.2: Top 20 most difference words between spam and ham domain

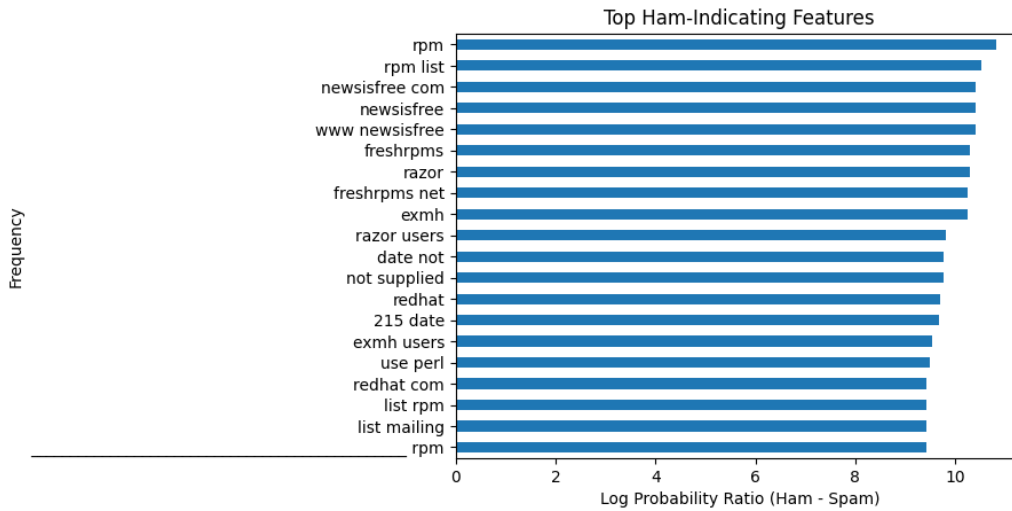


Figure 4.3: Top 20 most difference words between ham and spam domain

This analysis reveals lexical features most strongly associated with spam classification, providing interpretability to the model's decision-making process.

Feature importance analysis revealed that marketing-related terms (free, click, offer) serve as primary discriminative features for classification. The

error analysis identified specific challenges in handling short messages and technical content that could inform future model refinements.

Chapter 5

Discussions & Conclusions

5.1 Discussions

This investigation has addressed the development of a machine learning approach for discriminating between legitimate and unsolicited email communications. The primary contributions of this work include: development of a robust methodology for extracting and normalizing email content from diverse formats, identification of discriminative lexical features between spam and legitimate communications, implementation and evaluation of a Naive Bayes classification model for automated filtering and analysis of model performance characteristics and limitations

The Naive Bayes classification model demonstrated strong performance in the spam classification task and resulted in Cross-validation accuracy averaged 98.2%, Test set performance achieved 98% accuracy, Precision for spam identification reached 97% and Recall for spam identification reached 96%. These performance metrics suggest that the model is viable for real-world spam filtering applications, particularly in environments where high recall is prioritized to minimize exposure to potentially harmful communications.

Several limitations constrain the generalizability of the current investigation with potential temporal bias in collected communications, limited representation of certain communication types and absence of non-English content for multilingual evaluation

5.2 Conclusions

This investigation has demonstrated that Naive Bayes classification, coupled with appropriate text preprocessing and TF-IDF vectorization, provides effective discrimination between spam and legitimate email communications.

The approach achieves high accuracy while maintaining computational efficiency suitable for practical applications.

The analysis of discriminative features revealed that specific lexical patterns serve as reliable indicators of message legitimacy, reinforcing theoretical understanding of linguistic differences between solicited and unsolicited communications. However, the identified limitations in feature representation and model assumptions suggest opportunities for enhancement through more sophisticated approaches.

Future research directions, particularly in semantic feature engineering and deep learning approaches, offer substantial potential for advancing the state of the art in automated email filtering. These advancements could significantly enhance information security, productivity, and communication reliability in digital environments.